

Project Title – Smart Study Generator Agent

Domain of Project – [Education]

- **Problem Statement -**

Students today have access to a vast amount of learning resources, including lecture notes, textbooks, research papers, YouTube tutorials, and online courses. However, this information is scattered across multiple platforms, making it difficult to organize, understand, and study effectively. Traditional learning tools provide static notes and lack personalized guidance, making exam preparation time-consuming and less efficient.

- **Objective -**

Build an AI-powered Smart Study Generator Agent that personalizes and simplifies learning for students. The solution should:

- **Aggregate content from lecture notes, textbooks, research papers, and online resources into structured study material.**
- **Allow students to upload PDFs, notes, images, or voice queries and generate summaries, flashcards, quizzes, and concept maps.**
- **Create personalized study plans based on syllabus, learning progress, and exam schedules.**
- **Recommend important topics by analyzing previous question papers and study patterns.**
- **Provide a real-time dashboard to track study progress, identify weak areas, and forecast exam readiness.**
- **Use IBM watsonx.ai, IBM Granite Models, IBM LangFlow/Orchestrate, and RAG to deliver accurate and intelligent learning assistance.**

- **Proposed Solution -**

The proposed solution is an AI-powered Smart Study Generator Agent built using IBM watsonx.ai, IBM Granite Models, IBM LangFlow, IBM Cloud, and Retrieval-Augmented Generation (RAG). The system acts as an intelligent study assistant that helps students learn efficiently by organizing learning resources and providing personalized study guidance.

The solution combines class notes, textbooks, PDFs, research papers, and online learning materials into one platform. Using RAG, it retrieves relevant educational content and generates concise summaries, flashcards, concept explanations, and quizzes.

The system consists of four intelligent agents:

- **Knowledge Retrieval Agent – Retrieves information from notes, books, and uploaded documents using RAG.**
- **Study Planner Agent – Creates personalized study schedules based on syllabus, exam dates, and learning progress.**
- **Content Generator Agent – Generates summaries, flashcards, quizzes, and concept maps.**

- **Performance Analysis Agent** – Tracks learning progress, identifies weak topics, and recommends improvement strategies.

An interactive dashboard allows students to upload notes, ask questions, and monitor study progress in real time.